HexOct knitting: the 3C swatch

Stacking ABCABCA, 24 stitches by 43 rows

DICTO · knitting swatch of 31 October 2026
The idea is DICTO's: the knitted fabric is an exact edge-on slice through a stacking of balls, coloured by the pieces between the balls. Chart computed exactly with Claude (Anthropic), generator hexknit.

How to knit every swatch

- Stockinette. Stitches 1 and the last are edge stitches, worked in the chart colour, not slipped, so the edges show the bias steps.
- Colour changes: intarsia with small bobbins, twisting at every change; carry a colour along the back only where its next block is 1–2 stitches away.
- Every bias row keeps the stitch count: one decrease and one increase, each one stitch in from its edge. In the colour runs the increased and decreased stitches are counted as they finish; work each in the colour of its run.
- Bobble (MB), in the colour of the stitch. RS rows: (k1, yo, k1) into the stitch, turn, p3, turn, k3, turn, p3, turn, sl1-k2tog-psso. WS rows: (p1, yo, p1) into the stitch, turn, k3, turn, p3, turn, k3, turn, p3tog, and push it to the RS; if it sits badly, purl the stitch plain and sew a bobble or French knot there afterwards.
- Abbreviations: DG dark gold, LG light gold, DB dark blue, LB light blue. Runs are given in working order (RS from the last stitch to stitch 1, WS from stitch 1 to the last).

3C: ABCABCA (ccccccc), 24 stitches × 43 rows

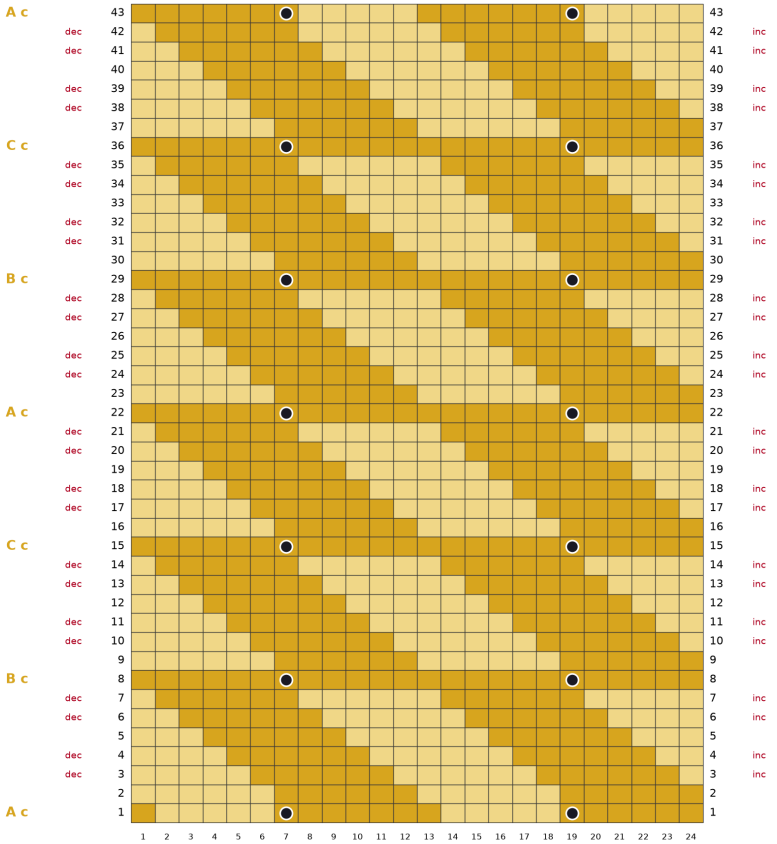
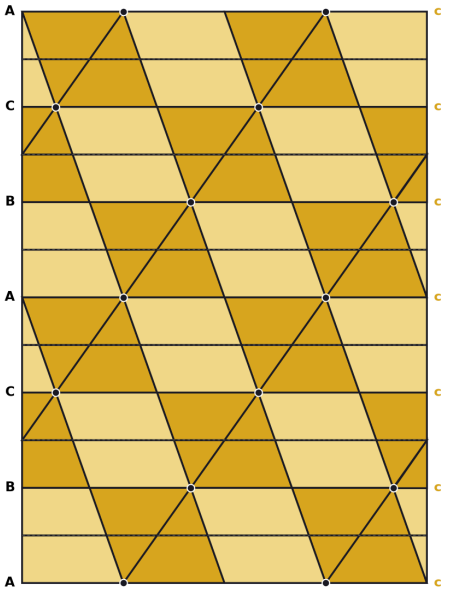


Chart. Seen from the RS, row 1 at the bottom; RS rows read right to left, WS rows left to right. Dots = bobbles. *dec*/*inc*: which edge of a bias row gets the decrease and which the increase.



The exact slice for this window (dots = ball centres; letters = layers; c gold, h blue; dark = tetrahedral, light = octahedral pieces).

DG dark gold: 580 sts · LG light gold: 452 sts. Total 1032 stitches.

Cast on 24 sts in DG; row 1 is a RS row. **Bias right** (decrease at the left edge, increase at the right, seen from the RS): rows 3, 4, 6, 7, 10, 11, 13, 14, 17, 18, 20, 21, 24, 25, 27, 28, 31, 32, 34, 35, 38, 39, 41, 42. **Bias left**: rows none. Net shift at the top: +24 stitches. **Bobbles** on RS rows 1, 15, 29, 43; on WS rows 8, 22, 36. Bind off in LG on the WS after row 43.

Row	Side	Colour runs (working order)	Bias (edge stitches)	Bobble
1	RS	6 DG, 5 LG, 7 DG, 5 LG, 1 DG		MB at st 7, 19
2	WS	6 LG, 6 DG, 6 LG, 6 DG		
3	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: k1, M1, … k2tog, k1	
4	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: p1, p2tog, … M1P, p1	
5	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
6	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: p1, p2tog, … M1P, p1	
7	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: k1, M1, … k2tog, k1	
8	WS	24 DG		MB at st 7, 19
9	RS	6 DG, 6 LG, 6 DG, 6 LG		
10	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: p1, p2tog, … M1P, p1	
11	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: k1, M1, … k2tog, k1	
12	WS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
13	RS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: k1, M1, … k2tog, k1	
14	WS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: p1, p2tog, … M1P, p1	
15	RS	24 DG		MB at st 7, 19
16	WS	6 LG, 6 DG, 6 LG, 6 DG		
17	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: k1, M1, … k2tog, k1	
18	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: p1, p2tog, … M1P, p1	
19	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
20	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: p1, p2tog, … M1P, p1	
21	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: k1, M1, … k2tog, k1	
22	WS	24 DG		MB at st 7, 19
23	RS	6 DG, 6 LG, 6 DG, 6 LG		
24	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: p1, p2tog, … M1P, p1	
25	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: k1, M1, … k2tog, k1	
26	WS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
27	RS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: k1, M1, … k2tog, k1	
28	WS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: p1, p2tog, … M1P, p1	
29	RS	24 DG		MB at st 7, 19
30	WS	6 LG, 6 DG, 6 LG, 6 DG		
31	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: k1, M1, … k2tog, k1	
32	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: p1, p2tog, … M1P, p1	
33	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
34	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: p1, p2tog, … M1P, p1	
35	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: k1, M1, … k2tog, k1	
36	WS	24 DG		MB at st 7, 19
37	RS	6 DG, 6 LG, 6 DG, 6 LG		
38	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: p1, p2tog, … M1P, p1	
39	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: k1, M1, … k2tog, k1	
40	WS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
41	RS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: k1, M1, … k2tog, k1	
42	WS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: p1, p2tog, … M1P, p1	
43	RS	5 LG, 7 DG, 5 LG, 7 DG		MB at st 7, 19